

Two-Step Quiz 2 Solutions

Step 1: Setup	Step 2: Execution
Question 1 $100\left(1 + \frac{.12}{4}\right)^{4n} = 200e^{.06n}$	Question 1 Answer: 408.5 Procedures: • Take logarithms of both sides • Apply log rules • Isolates n and solve (calculator)
Question 2 $A_1(3) - A_1(1) = A_2(3) - A_2(1)$ where $A_1(t) = 200(1 + .06t) \text{ and } A_2(t) = 100 + kt^2$	Question 2 Answer: 3 Procedures: • Isolates k and solve (calculator)
Question 3 $50(.95)^{-1} e^{\int_1^{n+1} \frac{2}{4+2t} dt} (1.05) = 100$	Question 3 Answer: 2.43 Procedures: • Integration and u-substitution • Apply log rules • Isolates n and solve (calculator)